

Supplementary Material for A Unified Kinematic Representation Enables Reusable Biological Joint Moment Estimation

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PARTICIPANT DEMOGRAPHICS

TABLE S1
PHYSICAL CHARACTERISTICS OF THE STUDY PARTICIPANTS.

Participant ID	Age	Gender	Height (m)	Mass (kg)
AB01	25	M	1.78	88.0
AB02	24	M	1.76	71.1
AB03	29	M	1.74	84.4
AB04	30	M	1.84	74.0
AB05	25	F	1.65	55.0
AB06	26	M	1.73	82.6
AB07	24	F	1.65	51.3
AB08	27	M	1.74	71.9

Eight healthy participants, including six males and two females, were recruited for this study. The cohort had a mean age of 26.3 ± 2.3 years, with a range of 24–30 years; a mean height of 1.74 ± 0.06 m, with a range of 1.65–1.84 m; and a mean body mass of 72.3 ± 13.3 kg, with a range of 51.3–88.0 kg. Detailed physical characteristics for each participant are provided in Table S1.

ROBOTIC HIP EXOSKELETON SPECIFICATIONS

A. Mechatronic and Physical Architecture

The bilateral robotic hip exoskeleton, shown in Fig. S1(a), delivers active sagittal-plane torque assistance at the hip joints. The electromechanical system comprises two brushless DC actuators with integrated motor encoders (AK80-9, CubeMars, China) powered by an onboard 22.2-V lithium-polymer (LiPo) battery. System-level computation is executed on an embedded microprocessor board (Jetson Orin Nano, NVIDIA, USA) paired with a custom printed circuit board (PCB) that manages low-latency sensor feedback and motor commands. The embedded inference and input/output architecture is illustrated in Fig. S1(c).

The structural frame features a 3D-printed nylon-based user interface supported by a flexible carbon-fiber joint that allows 5° of passive hip abduction and adduction in the frontal plane. The sagittal-plane range of motion provides up to 100° of hip flexion and 30° of hip extension. The total mass of the worn

system is 4.5 kg. Body fit is customized across users through four adjustable attachment locations: a pelvis band, upper and lower thigh orthotic shells, and an electronics backpack.

TABLE S2
ROBOTIC HIP EXOSKELETON HARDWARE SPECIFICATIONS.

Parameter	Specification / Component
Total mass	4.5 kg
Actuation	$2 \times$ brushless DC motors with encoders (CubeMars AK80-9)
Peak torque capability	18 Nm per joint
Power source	22.2-V LiPo battery
Embedded processing	Nvidia Jetson Orin Nano Devkit and custom sensor/actuator PCB
User interface frame	3D-printed nylon shell and flexible carbon-fiber frontal joint
Sagittal range of motion	100° flexion and 30° extension
Frontal range of motion	5° passive abduction/adduction
Attachment locations	Pelvis band, upper/lower thigh orthotics, and electronics backpack

ROBOTIC KNEE EXOSKELETON SPECIFICATIONS

A. Mechatronic and Physical Architecture

The bilateral robotic knee exoskeleton, shown in Fig. S1(b), delivers active sagittal-plane torque assistance at the knee joints; however, only one side of the device was used in this experiment. The electromechanical system comprises two brushless DC actuators with integrated motor encoders (AK80-9, CubeMars) powered by a shared onboard 22.2-V lithium-polymer battery. System-level computation is executed on an embedded microprocessor board (Jetson Orin Nano, NVIDIA, USA) paired with a custom PCB that manages low-latency sensor feedback and motor commands. The embedded inference and input/output architecture is illustrated in Fig. S1(c).

The structural frame features a fully 3D-printed user interface and linkage assembly. The sagittal-plane range of motion provides up to 120° of knee flexion and allows full knee extension (0°). The total mass of the worn system is 2.2 kg, comprising 0.9 kg for the waist belt and electronics and 1.3 kg for the unilateral knee attachment. Body fit is customized across users through adjustable thigh and shank orthotic shells and an electronics backpack.

WEARABLE KINEMATICS SYSTEM

A. IMU Sensor Configuration

Full-body kinematics were recorded using a wireless inertial motion capture system (MTw Awinda, Movella, Netherlands).

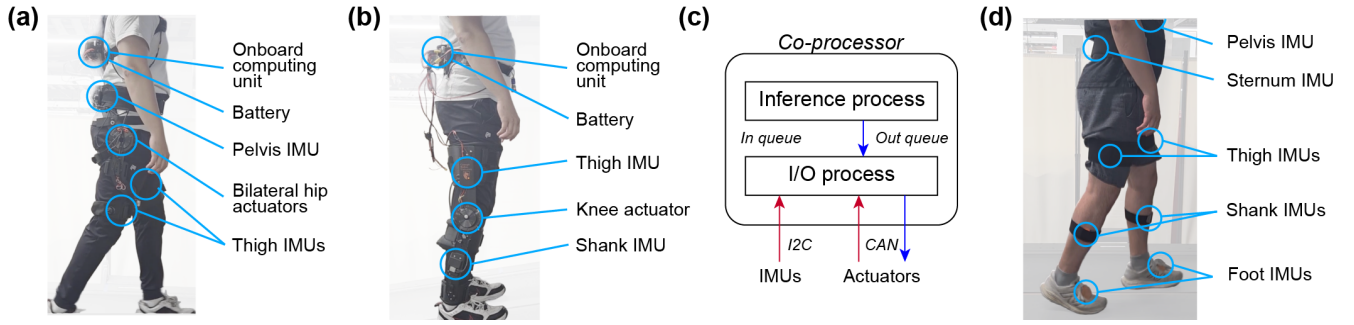


Fig. S1. Overview of the wearable systems and embedded computing architecture used in this study. (a) Bilateral robotic hip exoskeleton, including the onboard computing unit, battery, pelvis IMU, bilateral hip actuators, and thigh IMUs. (b) Robotic knee exoskeleton configuration used during data collection, including the onboard computing unit, battery, thigh IMU, knee actuator, and shank IMU. (c) Embedded co-processor architecture. Separate inference and input/output processes exchange data through input and output queues. IMU measurements are received through I²C, and actuator commands are transmitted through CAN. (d) Placement of the sternum, pelvis, bilateral thigh, bilateral shank, and bilateral foot IMUs in the Xsens Awinda sensor suite.

The experimental configuration, shown in Fig. S1(d), comprised seven inertial measurement units (IMUs) secured to the pelvis, bilateral thighs, bilateral shanks, and bilateral feet using adjustable straps.